Application of Derivatives

Rate of Measure :

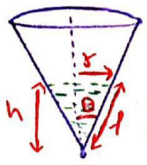
→ Rate is always w.r.t time (until specified something else).

→ Rate of one quantity will be given and rate of change of some other quantity is asked.

→ Always find a way to write 'Unknown quantity' in terms of 'Known Quantity' and then differentiate.

Ques A water tank has the shape of a right circular cone with axis vertical and vertex downwards. Its semi vertical angle is $\tan^{-1}(\frac{3}{4})$. Water is poured in it at a constant rate of $6\ m^3/hr$. The rate ($in\ m^2/hr$) at which wet curved surface area of the tank is increasing when the depth of water in tank is 4m is ___

Sol^n



Since, $\tan\theta = \frac{3}{4} = \frac{r}{h}$

$\Rightarrow \boxed{h = \frac{4r}{3}}$

We are given,

$\frac{dV}{dt} = 6\ m^3/hr \Rightarrow \frac{d}{dt}\left(\frac{1}{3}\pi r^2 h\right) = 6$

$\Rightarrow \frac{4}{9}\pi \frac{d(r^3)}{dt} = 6 \Rightarrow \frac{4\pi}{9}\cdot 3r^2 \cdot \frac{dr}{dt} = 6$

$\Rightarrow \boxed{\frac{dr}{dt} = \frac{9}{2\pi r^2}}$

We need → $\frac{d(\text{Curved surface area})}{dt}$

$\frac{d(CSA)}{dt} = \frac{d(\pi r l)}{dt}$ $\left[\because l^2 = r^2 + h^2,\ l^2 = \frac{25r^2}{9},\ \Rightarrow l = \frac{5r}{3}\right]$

$\frac{d(CSA)}{dt} = \frac{d}{dt}\left(\pi \cdot r \cdot \frac{5r}{3}\right)$

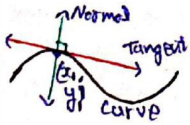
$= \frac{5\pi}{3}\frac{dr^2}{dt} \Rightarrow \frac{5\pi}{3}\cdot 2r \cdot \frac{dr}{dt}$

$= \frac{5}{10\pi r}\cdot \pi \cdot \frac{9}{2\pi r^2} \Rightarrow \frac{15}{r}$

We need $\frac{d(CSA)}{dt}\Big|_{h=4m}$ $\because h = \frac{4r}{3} = 4 \Rightarrow r = 3$

So, $\boxed{\frac{d(CSA)}{dt} = \frac{15}{3} = 5\ m^2/hr}$ Ans

Tangents and Normals :



Slope of tangent,

$\boxed{m_T = \frac{dy}{dx}\Big|_{(x_1,y_1)}}$

And, $\boxed{m_T \cdot m_N = -1}$

Equation of tangent

$\boxed{T : (y - y_1) = m_T(x - x_1)}$

Ques Let the tangent to the curve $x^2+2x-4y+9=0$ at the point P(1,3) on it meet y-axis at A. Let the line passing through P and parallel to the line $x-3y=6$ meet the parabola $y^2=4x$ at B. If B lies on the line $2x-3y=8$, then $(AB)^2$ is equal to ___.

Sol^n Slope of Tangent, ⇒

$2x + 2 - 4\frac{dy}{dx} = 0$ $[\because P \equiv (1,3)]$

$2(1) + 2 - 4\frac{dy}{dx} = 0 \Rightarrow \boxed{\frac{dy}{dx} = 1 = m_T}$

Eq^n of Tangent ⇒

$y - 3 = 1(x-1) \Rightarrow \boxed{y = x+2}$

$\boxed{A \equiv (0,2)}$

Now, for the line passing through P ⇒

P(1,3) $m_L = \frac{1}{3}$ (Bcz parallel to $y = \frac{x}{3} - 2$)

So, $L: (y-3) = \frac{1}{3}(x-1) \Rightarrow 3y - 9 = x - 1$

$L: \boxed{3y = x+8}$ → meets parabola $y^2=4x$

For P.O.I ⇒

$\left(\frac{x+8}{3}\right)^2 = 4x \Rightarrow x^2 + 64 + 16x = 36x$

$x^2 - 20x + 64 = 0 \rightarrow x = 16,\ y = 8$

$\rightarrow x = 4,\ y = 4$

Now, B also lies on $2x-3y=8$, so

$2(16) - 3(8) = 8$	$2(4) - 3(4) = 8$
$32 - 24 = 8$	$8 - 12 = 8$
$8 = 8$ (✓)	$-4 = 8$ (✗)

Hence, $\boxed{B \equiv (16,8)}$

So, $(AB)^2 = (16-0)^2 + (8-2)^2 = 256 + 36$

$\boxed{(AB)^2 = 292}$ Ans

Ques If the eqn of the normal to $y = \frac{x-a}{(x+b)(x-2)}$ at the point (1,-3) is $x - 4y = 13$ then the value of $(a+b)$ is ___.

Sol^n $m_N = \frac{1}{4} \Rightarrow \boxed{m_T = -4}$

So, $\frac{dy}{dx} = -4 = \frac{1(x+b)(x-2) - (2x-2+b)(x-a)}{(x+b)^2(x-2)^2}$

$-4 = \frac{(1+b)(1-2) - (2-2+b)(1-a)}{(1+b)^2(1-2)^2}$

$-4(1+b^2+2b) = -1 - b - b - ab$

$-4 - 4b^2 - 8b = -1 - 2b + b(-2-3b)$

[Since, (1,-3) lie on curve,

$-3 = \frac{1-a}{(1+b)(-1)} \Rightarrow -3 = \frac{1-a}{1+b}$

$\Rightarrow \boxed{a = -2 - 3b}$]

$-b^2 - 4b - 3 = 0 \Rightarrow b^2 + 4b + 3 = 0$

Bcz it makes $f(x)$ not defined @ (1,-3) ← $b = -1$ (✗), $b = -3$ (✓)

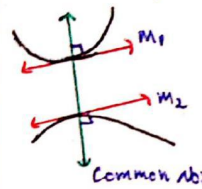
So, $a = -2 + 9 = 7$ $b = -3$

$\boxed{a + b = 4}$ Ans

Shortest Distance b/w two Curves :

→ Shortest distance b/w two curves is always along the 'common normal'.

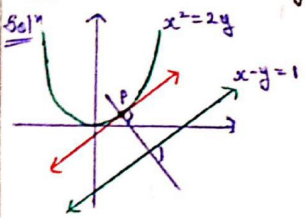
→ Means, the tangents for both curves would be parallel and with same slope



$\boxed{m_1 = m_2}$

Ques The shortest distance b/w the line $x-y=1$ and the curve $x^2 = 2y$ is ___.

Sol^n



Instead of calculating S.D. b/w parabola & line, we calculate S.D. b/w tangent & line as both are same.

For point $P(x_1, y_1)$

$x^2 = 2y$ ← m_T = slope of line sharing common normal

$2x = 2\left(\frac{dy}{dx}\right)$ $[m_T = 1]$

$2x_1 = 2(1) \Rightarrow \boxed{x_1 = 1}$ $\boxed{y_1 = 1/2}$

So, shortest distance of $P(1, 1/2)$ from $x-y=1$

$P = \left|\frac{1 - 1/2 - 1}{\sqrt{1^2 + (-1)^2}}\right| = \frac{1}{2\sqrt{2}}$ Ans

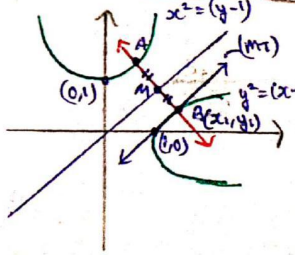
Ques Minimum distance b/w the curves $y^2 = x-1$ and $x^2 = y-1$ is equal to ___.

Sol^n

NOTE: Agar 2 curves jinme x aur y interchanged ho, woh $x=y$ ke across symmetric hote hai aur $x=y$ se equi-distant bhi.

So, we can calculate shortest distance as,

$\boxed{\text{S.D.} = 2(\text{Shortest distance of any one curve from } x=y)}$



m_T = slope of $x=y$

$\boxed{m_T = 1}$

Now,

$y^2 = x - 1$

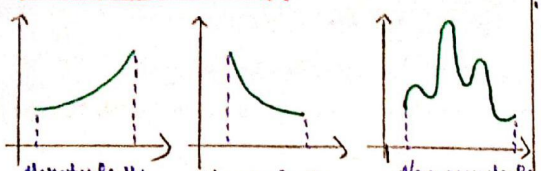
$2y\frac{dy}{dx} = 1$

$\boxed{y_1 = 1/2}$ $\boxed{x_1 = -5/4}$

So, $AM = \left|\frac{\frac{1}{2} - \frac{-5}{4}}{\sqrt{1^2 + 1^2}}\right| = \frac{3}{4\sqrt{2}}$

Now, S.D. = AB = 2(AM) ⇒ $\boxed{AB = \frac{3}{2\sqrt{2}}}$ Ans

Monotonic Functions :



Monotonically ↑sing function

$\frac{dy}{dx} \geq 0$

Monotonically ↓sing function

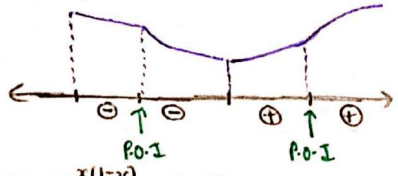
$\frac{dy}{dx} \leq 0$

Non-monotonic function

* $\left[\frac{dy}{dx} = 0\right]$ at points at which func^n is not defined are called critical points.

* Points across which monotonic nature of $f(x)$ remains same are called 'Point of Inflection' (P.O.I)

Eg;



Ques $f(x) = xe^{x(1-x)}$, $x \in R$.

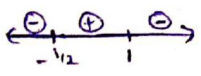
Soln $f(x) = xe^{x-x^2}$

$f'(x) = x \cdot e^{x-x^2}(1-2x) + e^{x-x^2} \cdot 1$

$= e^{x-x^2}(x - 2x^2 + 1)$

$= -(e^{x-x^2})(2x^2 - x - 1)$

e^{x-x^2} Always +ve; $(x-1)(2x+1)$



$f(x)$ ↑sing in $\left(-\frac{1}{2}, 1\right)$

$f(x)$ ↓sing in $(-\infty, -\frac{1}{2}) \cup (1, \infty)$

Ques Find the interval in which following $f(x)$ is ↑sing, defined on $R - \{-1, 1\}$

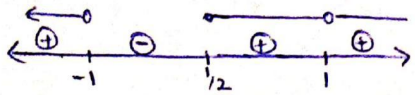
$$f(x) = 3\ln\left|\frac{x-1}{x+1}\right| - \frac{2}{x-1}$$

Soln

$$f'(x) = 3\left(\frac{x+1}{x-1}\right)\left[\frac{1(x+1) - 1(x-1)}{(x+1)^2}\right] + \frac{2}{(x-1)^2}$$

$$= \frac{3(2)}{(x-1)(x+1)} + \frac{2}{(x-1)^2} \Rightarrow \frac{6(x-1) + 2(x+1)}{(x-1)^2(x+1)}$$

$$f'(x) = \frac{8x-4}{(x-1)^2(x+1)} \Rightarrow \frac{4(2x-1)}{(x-1)^2(x+1)}$$



$f(x)$ is ↑sing in $(-\infty, -1) \cup \left[\frac{1}{2}, \infty\right) - \{1\}$ Ans

Ques Let $f(x) = 2x + \tan^{-1}x$ and $g(x) = \ln(\sqrt{1+x^2} + x)$, $x \in [0,3]$. Then

(A) There exist $x \in [0,3]$ such that $f'(x) < g'(x)$

(B) max. $f(x)$ > max. $g(x)$

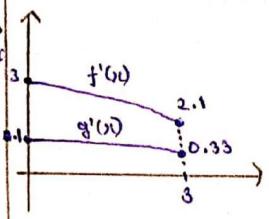
(C) There exist $0 < x_1 < x_2 < 3$ such that $f(x) < g(x)$

(D) min. $f'(x)$ = 1 + max. $g'(x)$

Sol^n $f'(x) = 2 + \frac{1}{1+x^2}$; $g'(x) = \frac{1}{\sqrt{1+x^2}}$

$f'(x)$: Min. (0,3), Max. (3, 2.1)

$g'(x)$: Min. (0,1), Max. $\left(3, \frac{1}{\text{S.m}}\right)$ ⇓ (3, 0.33)



From the graph of $f'(x)$ & $g'(x)$, we can clearly deduce

(A) No such $x \in [0,3]$ for which $f'(x) < g'(x)$, $f'(x) > g'(x)$ always.

(B) Max. $f'(x) = 3$; Max. $g'(x) = 1$ ⇒ $3 > 1$ (correct option)

(C) There does not exist any point, for which $f(x) < g(x)$.

(D) 1 + max. $g'(x) = 1 + 1 = 2$

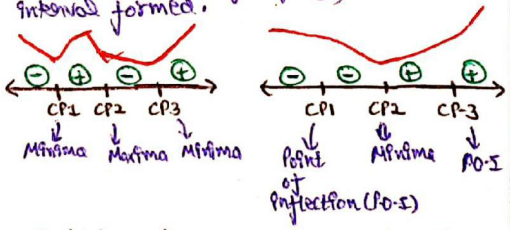
Min. $f'(x) = 2.1$ $[2 \neq 2.1]$

Maxima and Minima :

→ Plot the points on number line for which

$$\frac{dy}{dx} = 0 / \text{Not defined}$$

→ Then check the sign of $\left(\frac{dy}{dx}\right)$ in each interval formed.



Ques If total maximum value of the function $f(x) = \left(\frac{\sqrt{3}e}{2\sin x}\right)^{\sin^2 x}$, $x \in \left(0, \frac{\pi}{2}\right)$ is $\left(\frac{k}{e}\right)$ then $\left(\frac{k}{e}\right)^8 + \frac{k^8}{e^5} + k^8$ is equal to ——.

Sol^n $y = \left(\frac{\sqrt{3}e}{2\sin x}\right)^{\sin^2 x} \Rightarrow \ln y = \sin^2 x \cdot \ln\left(\frac{\sqrt{3}e}{2\sin x}\right)$

$$\frac{y'}{y} = \sin^2 x\left(\frac{2\sin x}{\sqrt{3}e}\right)\left(\frac{-2\cos x(\sqrt{3}e)}{4\sin^2 x}\right) + \ln\left(\frac{\sqrt{3}e}{2\sin x}\right) \times 2\sin x\cos x$$

$$\frac{y'}{y} = \frac{-\sin x \cdot \cos x \cdot 2}{2} + \ln\left(\frac{\sqrt{3}e}{2\sin x}\right) \cdot 2\sin x \cdot \cos x$$

$$\frac{y'}{y} = \sin 2x\left[-\frac{1}{2} + \ln\left(\frac{\sqrt{3}e}{2\sin x}\right)\right]$$

$$y' = y \cdot \sin 2x \cdot \left(-\frac{1}{2} + \ln\left(\frac{\sqrt{3}e}{2\sin x}\right)\right) = 0$$

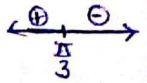
Can't be zero. Exp. $f(x)$ can have min. value 1

$x \in (0, \pi/2)$ ⇒ $2x \in (0, \pi)$ ⇒ $\sin 2x \in (0, 1]$ ⇒ Can't be zero

Means y' can be zero only when this part is 0

$$\Rightarrow \ln\left(\frac{\sqrt{3}e}{2\sin x}\right) = \frac{1}{2} \Rightarrow \frac{\sqrt{3}e}{2\sin x} = \sqrt{e}$$

$$\Rightarrow \sin x = \frac{\sqrt{3}}{2} \text{ i.e. } \boxed{x = \pi/3}$$



$$y_{max} = \left(\frac{\sqrt{3}e}{2\sin x}\right)^{(\sqrt{3}/2)^2} \Rightarrow \left(\frac{\sqrt{3}e}{\sqrt{3}}\right)^{3/4}$$

($\sin x = \sqrt{3}/2$)

$\boxed{y_{max} = e^{3/8}}$ ⇒ $k = e \cdot e^{3/8}$

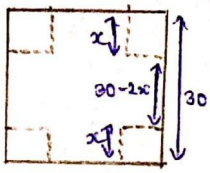
$\boxed{k = e^{11/8}}$

Hence,

$$\boxed{\left(\frac{k}{e}\right)^8 + \frac{k^8}{e^5} + k^8 = e^3 + e^6 + e^{11}} \text{ Ans}$$

Ques A square piece of tin of side 30 cm is to be made into a box w/o top by cutting a square from each corner & fold to form box. If vol. of box is max., it surface area is ——.

Soln



For the box,

$l = b = 30 - 2x$

$h = x$

$V = x(30-2x)^2$

$V = 4(225x + x^3 - 30x^2)$

Now, $\frac{dV}{dx} = 4(225 + 3x^2 - 60x) = 0$

$75 + x^2 - 20x = 0$

$(x-5)(x-15) = 0$

$x = 5$ (✓), 15 (✗) → Because $30 - 2(15) = 0$ Not possible

Hence,

Max. area $= (30-2(5))^2 + 4(30-2(5)) \times 5$

$= 20^2 + 4(20)(5) \Rightarrow$ 800 sq-cm Ans

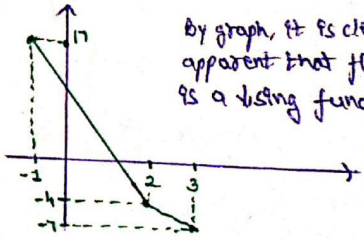
Ques The sum of the absolute max. & min. value of $f(x) = |x^2 - 5x + 6| - 3x + 2$ in the interval $[-1, 3]$ is equal to ——.

Sol^n $f(x) = |(x-2)(x-3)| - 3x + 2$

$f(2) = -4$

$f(3) = -7$

$f(-1) = 17$



By graph, it is clearly apparent that $f(x)$ is a ↓sing function

So $(f(x))_{max} = 17$ $(f(x))_{min} = -7$

$\boxed{\text{Max.} + \text{Min.} = 10}$ Ans

Ques A wire of length 20 m is to be cut into two pieces. Piece of length l_1 is bent to make a square of area A_1 and other piece of length l_2 is made into circle of area A_2. If $(2A_1 + 3A_2)$ is min. then $(\pi l_1) : (l_2)$ is equal to ——.

Soln

side $= \frac{l_1}{4}$; rad. $= \frac{l_2}{2\pi}$

$2A_1 + 3A_2 \geq 2\left(\frac{l_1}{4}\right)^2 + 3\pi\left(\frac{l_2}{2\pi}\right)^2$

$\Rightarrow \frac{l_1^2}{8} + \frac{3l_2^2}{4\pi}$

$\frac{2l_1 l_1'}{8} + \frac{6l_2 l_2'}{4\pi} = 0$

[$\because l_1 + l_2 = 20$

$l_1' + l_2' = 0$

$\Rightarrow l_2' = -l_1'$]

$\frac{l_1 l_1'}{4} - \frac{6l_2 l_1'}{4\pi} = 0$

$\Rightarrow \boxed{\pi l_1 / l_2 = 6}$ Ans